Dataset1 7-feature formal report

Performance and SHAP interpretation for the 7-feature design: pH, UV254, temperature, TOC, bromide, Cl2 dose, and contact time.

Final 7-feature highlights

- Common feature backbone retained for THM4, BDCM, and DBCM.
- Final model choices: Random Forest for THM4 and BDCM, MLP for DBCM.
- Complete-case sample count: 488 rows (382 train / 106 test).

Generated on 2026-04-01

Executive summary

This report consolidates the final 7-feature formal runs and the target-specific SHAP interpretation built from the winning model of each target.

All three targets were analyzed with the same 7-feature backbone. The final recommendation is target-specific at the model level, but unified at the feature-set level: the 7-feature design is retained for THM4, BDCM, and DBCM.

- THM4: Random Forest reached R2 = 0.8550 and delivered the best overall fit.
- BDCM: Random Forest reached R2 = 0.8432 and remained the most stable formal winner.
- DBCM: MLP reached R2 = 0.7134 and clearly outperformed the 7-feature XGBoost and KAN candidates.
- Across SHAP results, Cl2 dose is consistently important, while contact time adds useful but secondary explanatory signal.

Experiment setup

The same 7-feature representation was used for all targets. Missing rows were removed only when one of the selected features or the current target was unavailable.

Feature set: pH, UV254, Temperature, TOC, Bromide, Cl2 dose, and Contact time.

Complete-case sample count for the 7-feature setting: 488 rows total (382 train / 106 test).

Formal tuning used 5-fold cross-validation with stability-aware selection. Each target was optimized independently because the best-performing model family differed across THM4, BDCM, and DBCM.

Final recommendations

Target	Winning model	RMSE	MAE	R2
THM4	Random Forest	37.378	25.934	0.8550
BDCM	Random Forest	8.430	6.321	0.8432
DBCM	MLP	4.924	3.338	0.7134

Formal results

The first table reports the complete formal test-set metrics under the 7-feature configuration. The second table compares each final winner against its strongest earlier reference setting.

All formal metrics under the 7-feature setting

Target	Model	RMSE	MAE	R2
THM4	Random Forest	37.378	25.934	0.8550
THM4	XGBoost	49.215	30.812	0.7487
THM4	MLP	42.686	31.260	0.8109
THM4	KAN	52.503	34.455	0.7140
BDCM	Random Forest	8.430	6.321	0.8432
BDCM	XGBoost	9.350	6.429	0.8071
BDCM	MLP	9.984	8.048	0.7801
BDCM	KAN	9.028	6.503	0.8202
DBCM	Random Forest	6.725	5.184	0.4655
DBCM	XGBoost	5.570	4.123	0.6333
DBCM	MLP	4.924	3.338	0.7134
DBCM	KAN	6.504	4.728	0.4999

Improvement versus the strongest earlier reference

Target	Prior best	Prior R2	7f R2	Delta R2	Delta RMSE
THM4	6f + Cl2 scout	0.7341	0.8550	+0.1210	+18.669
BDCM	6f + Cl2 scout	0.7274	0.8432	+0.1158	-13.445
DBCM	6f + contact formal	0.6355	0.7134	+0.0779	-0.629

Performance interpretation

- Random Forest remained the most reliable family for THM4 and BDCM, which suggests that both targets benefit from robust non-linear partitioning under the expanded 7-feature design.
- DBCM behaved differently: its best formal result came from MLP rather than tree-based models, indicating a more distributed and smoother response surface across the selected predictors.
- KAN did not win any of the three final formal runs, so the final 7-feature recommendation now favors simpler and more stable model families for deployment or reporting.

SHAP overview

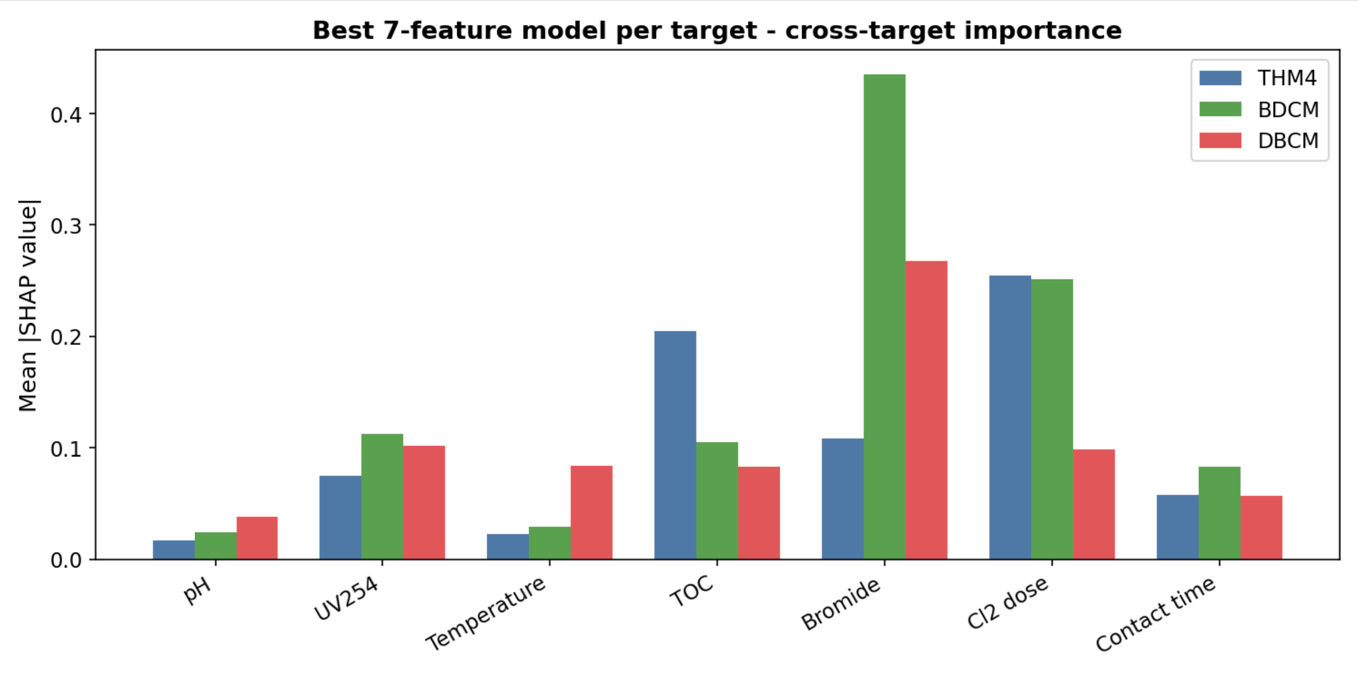
The overview below combines the cross-target feature-importance picture with the most important model-selection takeaways.

- THM4 benefited the most from Cl2 dose. The 7-feature Random Forest improved over the earlier 6-feature + Cl2 scout winner from R2 = 0.8275 to R2 = 0.8550.

- BDCM also preferred the 7-feature setting, with Random Forest reaching R2 = 0.8432 and slightly exceeding the earlier best 6-feature + Cl2 scout result (R2 = 0.8270).

- DBCM showed the clearest gain from fully unifying around 7 features: the final MLP achieved R2 = 0.7134, improving over the previous 6-feature + contact-time formal MLP result of R2 = 0.6355.

- The 7-feature DBCM run is especially important because scout had favored XGBoost, but formal tuning shifted the final winner to MLP, indicating better generalization after full validation.



Cross-target mean absolute SHAP values for the three selected 7-feature models.

Top SHAP ranks under the selected 7-feature models

Target	Top 1	Top 2	Top 3	Contact rank
THM4	Cl2 dose	TOC	Bromide	5
BDCM	Bromide	Cl2 dose	UV254	5
DBCM	Bromide	UV254	Cl2 dose	6

- Final recommendation: retain 7 features as the common dataset1 design backbone.

- Model choice remains target-specific: Random Forest for THM4 and BDCM, MLP for DBCM.

- Cl2 dose is the strongest newly added explanatory variable across the full 7-feature study.

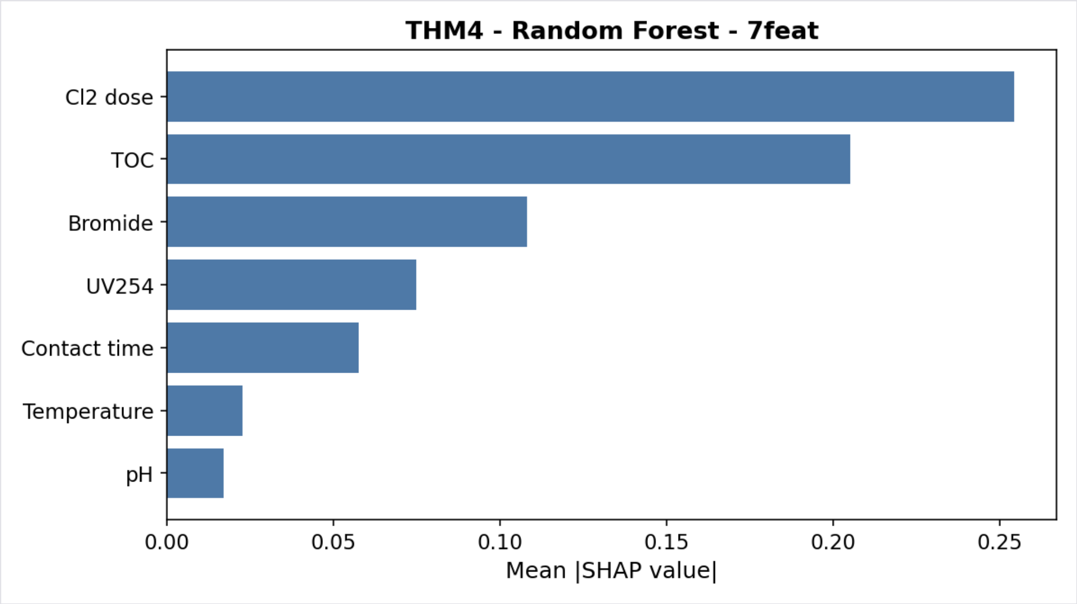
- Contact time is consistently non-zero in SHAP, but it is not a top-3 driver for any of the three targets.

THM4 interpretation

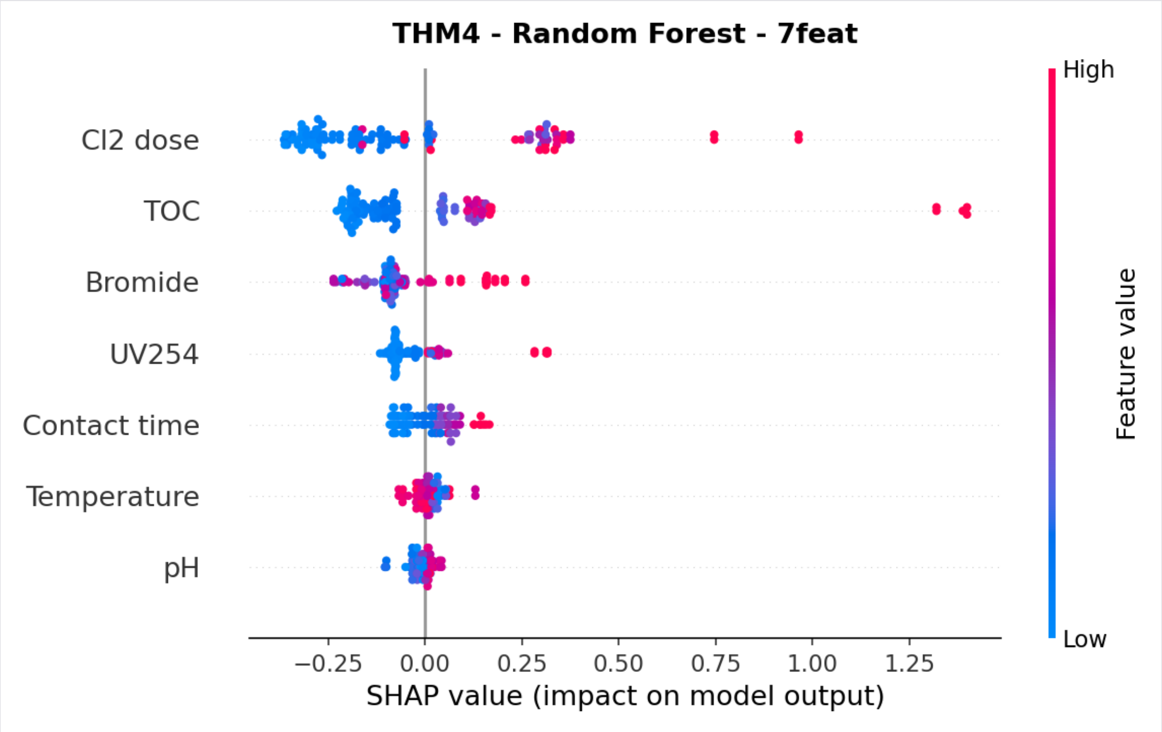
Best model: Random Forest. This page combines the ranking of mean absolute SHAP values with the SHAP beeswarm pattern.

Top SHAP drivers for THM4: Cl2 dose (0.254), TOC (0.205), Bromide (0.108).

Interpretation: THM4 is primarily driven by Cl2 dose and TOC, while bromide provides a secondary but still meaningful contribution. Contact time is helpful, but it is not the dominant source of variance.



THM4: mean absolute SHAP values.



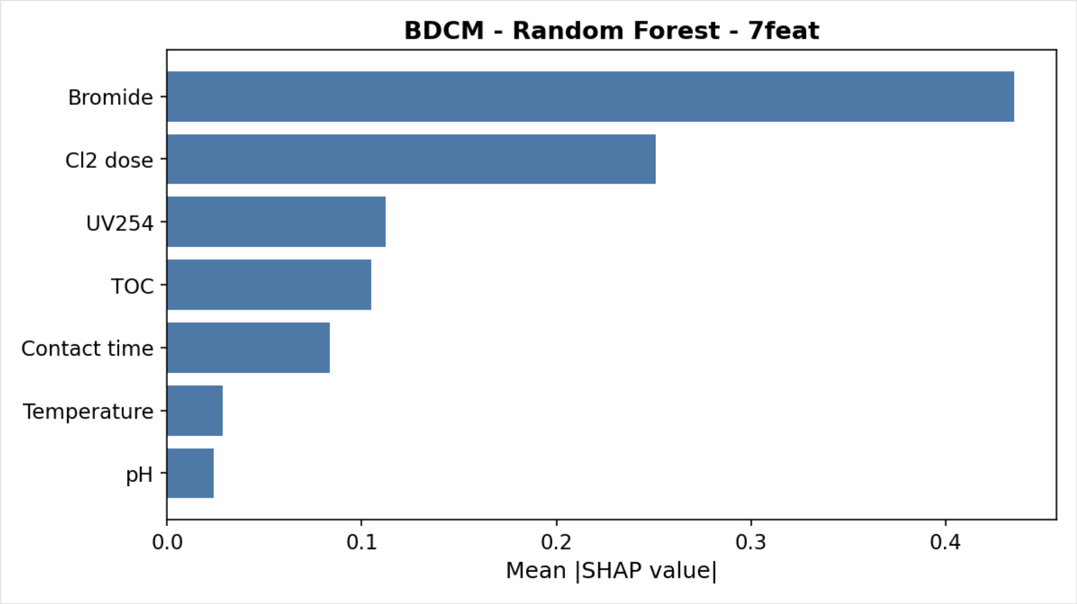
THM4: SHAP beeswarm summary.

BDCM interpretation

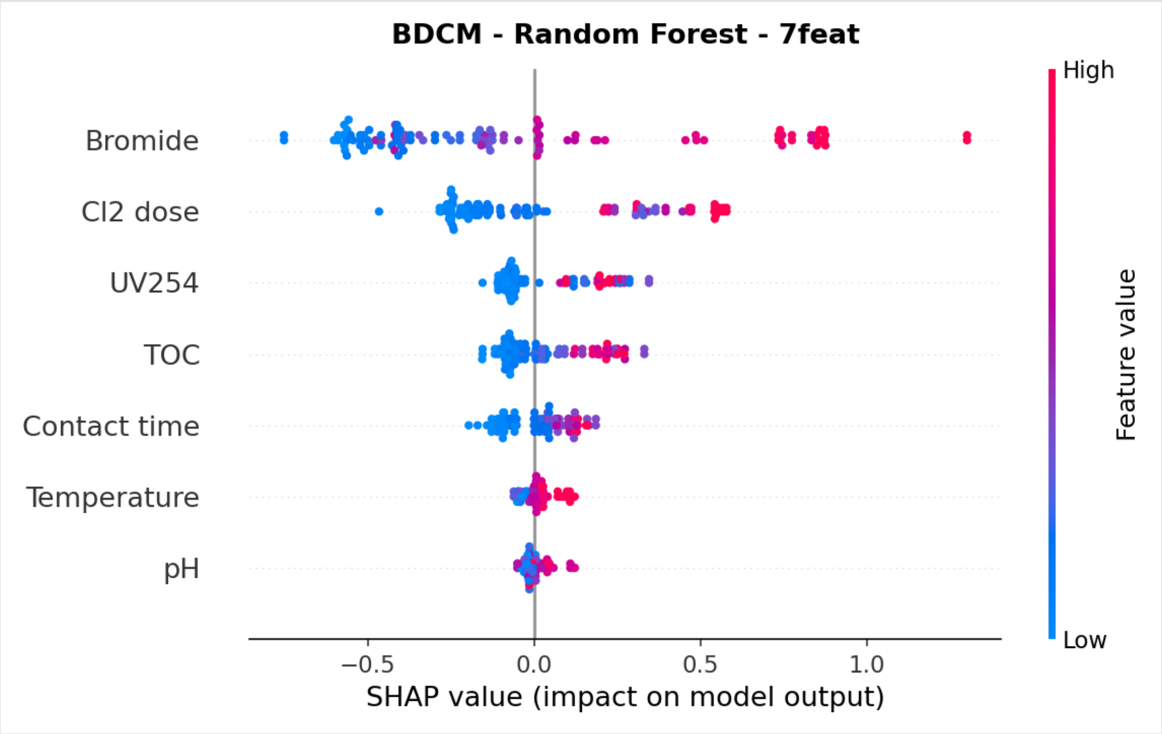
Best model: Random Forest. This page combines the ranking of mean absolute SHAP values with the SHAP beeswarm pattern.

Top SHAP drivers for BDCM: Bromide (0.436), Cl2 dose (0.251), UV254 (0.112).

Interpretation: BDCM is bromide-led, with Cl2 dose providing the second-largest incremental effect. UV254 and TOC stay relevant, which suggests that precursor quality still matters after disinfection-related variables are included.



BDCM: mean absolute SHAP values.



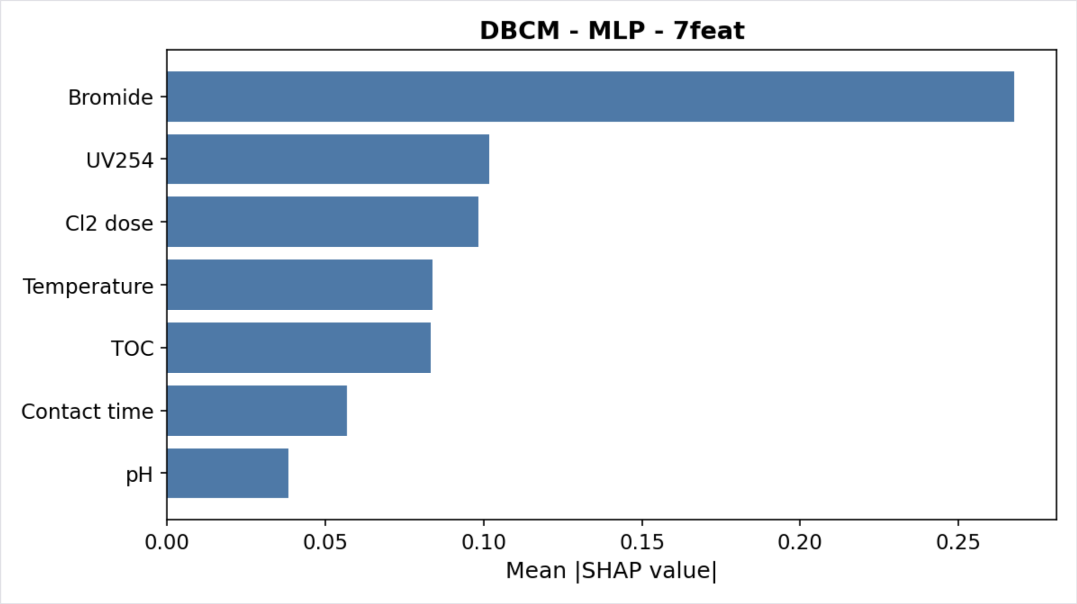
BDCM: SHAP beeswarm summary.

DBCM interpretation

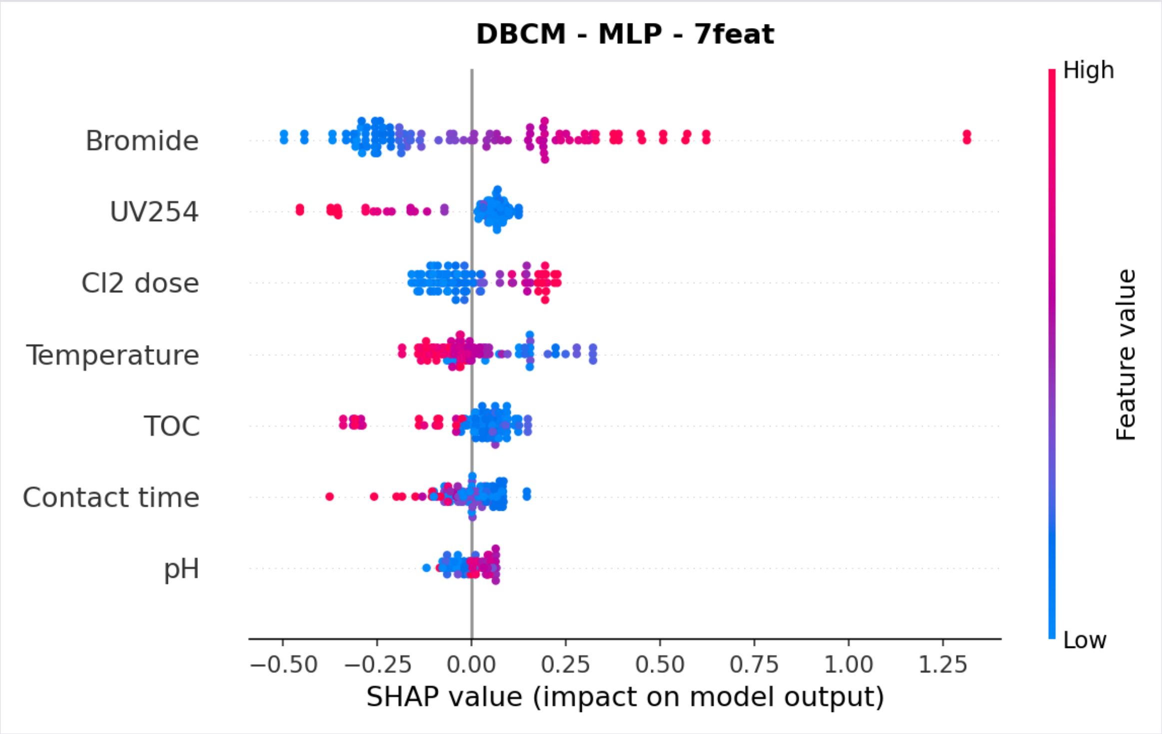
Best model: MLP. This page combines the ranking of mean absolute SHAP values with the SHAP beeswarm pattern.

Top SHAP drivers for DBCM: Bromide (0.268), UV254 (0.102), Cl2 dose (0.098).

Interpretation: DBCM is more evenly distributed across bromide, UV254, Cl2 dose, temperature, and TOC than the other targets. Contact time contributes positively, but the strongest 7-feature DBCM signal still comes from bromide and precursor-related variables.



DBCM: mean absolute SHAP values.



DBCM: SHAP beeswarm summary.